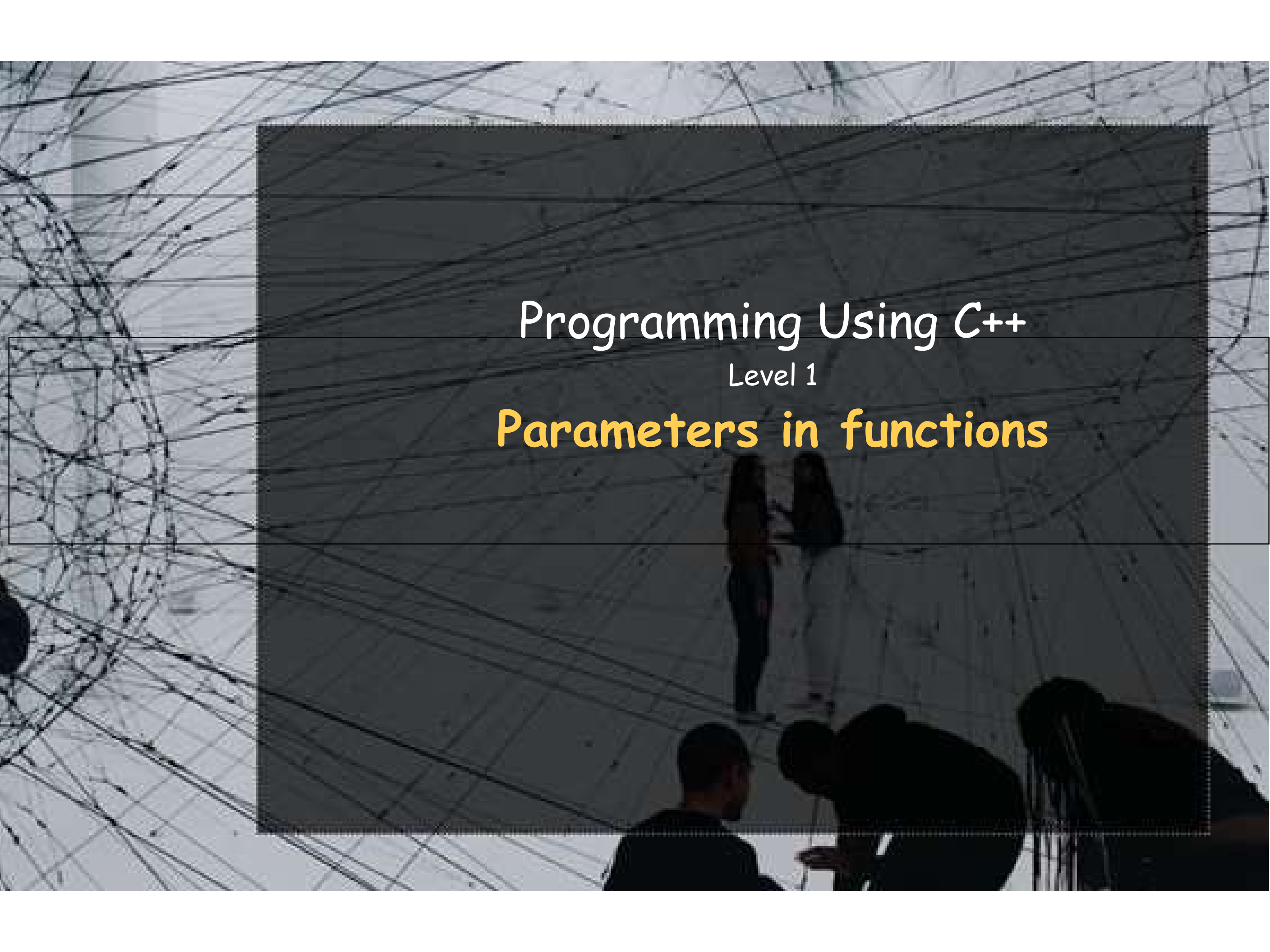


Programming Using C++ Level 1

Parameters in functions



Programming Using C++

Level 1

Parameters in functions

Definition

- ◆ A parameter is a variable that allows a function to receive data from the code that calls it.
- ◆ Parameters are written inside the function parentheses.
- ◆ The values passed to the function when it is called are called arguments.

Explanation

- ◆ This program defines `MySumFunction()` with two parameters: `Num1` and `Num2`.
- ◆ The function receives two numbers, adds them together, and returns the result.
- ◆ The values entered by the user are passed to the function when `MySumFunction(Num1, Num2)` is called.

Steps in main()

1 Define the function

- ◆ Create a function that accepts two integer parameters.

2 Receive the values

- ◆ The parameters Num1 and Num2 receive the values passed to the function.

3 Perform the operation

- ◆ Add Num1 and Num2 together.

4 Return the result

- ◆ Use return to send the calculated sum back to the calling code.

5 Call the function

- ◆ Pass the two variables to MySumFunction().

6 Display the result

- ◆ Print the returned value using cout.

💡 Example

◆ Function definition:

```
int MySumFunction(int Num1, int Num2)
{
    return Num1 + Num2;
}
```

◆ Function call:

```
MySumFunction(10, 20)
```

◆ The function receives 10 and 20 and returns 30.

➡ Result

◆ If the user enters 10 and 20, the program displays:
The sum is: 30

Note

- ◆ Parameters are defined in the function declaration, for example: `int Num1, int Num2`.
- ◆ Arguments are the actual values or variables passed when calling the function.
- ◆ `return` sends a value back to the code that called the function.
- ◆ The return type `int` means the function returns an integer value.

⊘ Warning

- ◆ The number and types of arguments should match the function's parameters.



Comparison

▨ Function with Parameters

- ✓ Can receive data from outside the function.
- ✓ Can work with different values each time it is called.
- ✓ Makes functions more flexible and reusable.

▤ Function without Parameters

- ✓ Does not receive values from the caller.
- ✓ Works only with data already available inside the function.
- ✓ May be less flexible for operations that need changing input.

✦✦ Summary

- ✓ Parameters allow functions to receive data.
- ✓ Arguments are the actual values passed to a function.
- ✓ Parameters make functions more flexible and reusable.

Thank you 😊